



Original research

Neurocognitive function influences dynamic postural stability strategies in healthy collegiate athletes



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ABSTRACT

Objectives: Poorer neurocognitive performance may increase lower extremity injury risk due to alterations in biomechanics. However, it is unclear if poorer neurocognitive function may be associated with altered dynamic postural stability. Therefore, the purpose of this study was to investigate the relationship between neurocognitive performance and dynamic postural stability in healthy collegiate athletes.

Design: Cross-sectional cohort.

Methods: Forty-five Division-I collegiate athletes (21 males, 24 females; age: 19.69 ± 1.50) completed neurocognitive assessments from the NIH Toolbox® (NIHTB). Three groups were established from the NIHTB composite score: high performers (HP), moderate performers (MP), and low performers (LP). Additionally, participants completed a dynamic hop-to-stabilization task. Accelerometer and gyroscopic data were recorded during landing through an inertial measurement unit (IMU) on the participant's low back. The root mean squared (RMS) of the accelerometer and gyroscope was calculated for the orthogonal planes and the resultant vector. Group differences for demographic variables, NIHTB composite scores, and IMU based measures were analyzed with one-way ANOVAs with Bonferroni post hoc analyses were performed. Cohen's d effect sizes were also calculated.

Results: Post hoc tests determined the LP group had higher vertical acceleration RMS values ($p = 0.013$, $d = -0.85$) and lower anteroposterior acceleration RMS values ($p = 0.005$, $d = 0.95$) compared to the HP group.

Conclusions: Neurocognitive performance may influence dynamic postural stability strategies in athletes. Higher neurocognitive performers may use different approaches to perform difficult postural tasks by adopting strategies associated with lower vertical and higher anteroposterior acceleration compared to lower neurocognitive performers.

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Practical implications

- Neurocognition may be related to dynamic postural control.
- Low neurocognitive performers demonstrated higher vertical and lower anteroposterior acceleration compared to high performers.
- Moderate neurocognitive performers utilize similar strategies as both low and high neurocognitive performers.

1. Introduction

Lower extremity injuries accounted for over 50% of injuries occurring in 15 different collegiate sports over a span of 16 years in the

United States.¹ Ankle sprains accounted for 14.8% of all reported injuries, while anterior cruciate ligament (ACL) injuries accounted for approximately 3% of all injuries.¹ Immediate consequences as a result of lower extremity injury may include, but are not limited to, pain, swelling, laxity and instability, functional limitations, decreased strength, range of motion deficits, and loss of proprioception.^{2,3} In addition to the immediate consequences of lower extremity injuries, the long-term consequences of these injuries are becoming increasingly apparent. Up to 75% of ankle sprain patients experienced symptoms such as weakness, swelling, pain, and instability 1.5 to 4 years following injury.⁴ Similarly, many soccer players who sustained ACL injuries, exhibited knee osteoarthritis, pain, and functional limitations 12 to 14 years following injury.^{5,6} The immediate and long-term consequences of these injuries underscores the importance of identifying the underlying causes of injury to develop effective prevention and rehabilitation strategies.

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Different aspects of physical performance, such as muscle strength, flexibility, and balance, amongst other factors have been examined as potential risk factors for lower extremity injury with mixed results.⁷ However, neurocognition has been explored as a potential influencing factor in lower extremity injury risk. For example, poorer reaction time, processing speed, and visual spatial disorientation have been identified in athletes who later sustained ACL injuries.⁸ Similarly, in a cohort of football players, poorer reaction times were associated with double the injury risk compared to those with faster reaction time.⁹ Though, poorer neurocognitive function has been associated with an increased lower extremity injury risk, the underlying mechanisms are unclear.

The heightened injury risk associated with poorer neurocognitive performance may be related to deviations in lower extremity biomechanics.^{10–12} This suggests that the ability to rapidly make decisions in complex environments may influence the way individuals perform multifaceted sport-related tasks.¹³ This is supported by studies which have identified individuals with poorer neurocognitive performance demonstrating greater vertical ground reaction forces, peak knee valgus angle, or peak external knee valgus moment during tasks such as drop jump landing or cutting maneuvers.^{10–12} Despite methodological differences across the studies, including the sport-related task, neurocognitive assessment, type of sports participation, and sex of the participants; individuals with poorer neurocognitive performance presented with biomechanical patterns that may predispose them to injury. Therefore, the link between neurocognitive performance and athletic movement patterns should be further investigated as a potential mechanism for lower extremity injury.

Exploring the relationship between neurocognition and other aspects of physical performance may provide important information to further understand how neurocognition is related to injury risk. Dynamic postural stability has been commonly examined as a factor for lower extremity injury risk and may have relevance in this context.^{14,15} Dynamic postural stability is seen in athletics as individuals are required to maintain balance following abrupt cutting and jumping tasks, a necessary component for greater success in sports such as basketball, football, and soccer.¹⁶ Previous studies have identified that dual-tasking (concurrently performing a cognitive and motor task) has a negative effect on dynamic postural stability.^{17–19} While these studies support the concept that neurocognition has an influence on dynamic postural stability, the relationship between baseline neurocognition and dynamic postural stability is less understood. Examining the relationship between baseline neurocognitive performance and dynamic postural stability may provide insight into the ability to optimize postural motor commands and further elucidate potential mechanics for injury risk. Therefore, the purpose of this study was to determine if differences in dynamic postural stability exist between healthy collegiate athletes with varying levels of baseline neurocognitive performance. We hypothesized that the athlete group with poorer neurocognitive function would exhibit greater acceleration and rotational velocity following landing compared to the groups with higher neurocognitive function.

2. Methods

Participants were recruited from the men's and women's soccer and cheerleading teams from a Division-I university. Participants were active on the roster and cleared for full participation at the time of testing. Participants were excluded if they: 1) sustained a concussion within the past 12 months, 2) had any lower extremity surgeries, 3) sustained a lower extremity injury within the past 3 months, or 4) possessed cardiovascular, neurological, respiratory, or other conditions that hindered full participation. Written informed consent was obtained from each participant prior to data collection in compliance with the University's institutional review board.

All participants completed data collection during a single visit to the research laboratory. After completing enrollment procedures,

participants completed a demographics, injury and medical history questionnaire to acquire information related to lower extremity musculoskeletal injury and concussion history. Participants then completed a battery of neurocognitive assessments and a dynamic hop-to-stabilization task on each limb.

A battery of four neurocognitive assessments were administered through the NIH Toolbox (NIHTB)® for the Assessment of Neurological and Behavioral Function. The assessments included the Flanker Inhibitory Control and Attention Test, Dimensional Change Card Sort Test, Pattern Comparison Processing Speed Test, and Picture Sequence Memory Test (Table 1).²⁰ All tests were completed on a 10.2" tablet (Apple iPad, 5th Generation; Apple, Cupertino, CA) with the participant seated at a table. All assessments were administered in a manner consistent with the instructional videos and standard operating procedures manual created for the NIHTB.^{20,21} These tests have been found to have good to excellent test-retest reliability (ICC) and moderate to good convergent validity (r, correlation to well-established "gold standards") for Dimensional Change Card Sort Test (0.81; 0.55), Flanker (0.83; 0.52), Pattern Comparison Processing Speed (0.73; 0.50–0.54), and Picture Sequence Memory Test (0.77; 0.64–0.72), respectively.^{22–24}

Fully-corrected standard scores (T-score) were calculated by the NIHTB application for all neurocognitive tests. These scores accounted for age, sex, education level, race, and ethnicity. Fully-corrected standard scores were based on a population norm of 50 ± 10 with higher scores representing greater neurocognitive function. For this study, the fully-corrected scores were summed from each NIHTB test to compute a composite score. From the percentile ranks of the composite scores, three independent groups of nearly equal size to represent high performers (HP [NIHTB Composite Score = 251.75 ± 16.04]), moderate performers (MP [NIHTB Composite Score = 220.00 ± 6.53]), and low performers (LP [NIHTB Composite Score = 183.73 ± 16.54]) (Table 2).^{11,12}

Dynamic postural stability was assessed using a hop-to-stabilization task.²⁵ This task involved a forward hop to a target area that was at a distance equal to 40% of the participant's height. Participants were instructed to begin in a double-limb stance then hop with both limbs over the 30.5 cm hurdle and land on a single limb in the target area. Participants were instructed to stabilize as quickly as possible and maintain single limb stance for a minimum of 3 s. Each participant completed three practice trials on each limb, followed by three successful trials for each limb. Trials were completed back-to-back on the same limb before proceeding to the opposing limb. Trials were considered unsuccessful and repeated if the participant failed to clear the hurdle, touched down with the opposite foot, landed outside the target area, or pivoted the foot during landing.

Linear acceleration and rotational velocity were recorded during the landing and stabilization phase of the hop-to-stabilization task through a single inertial measurement unit (IMU) (Blue Thunder v1.5, iMeasureU Ltd., Auckland, New Zealand) placed on the participant's lower back (spinous process of L4). The IMU was oriented with the positive Y-axis oriented superiorly and the positive Z-axis oriented posteriorly and secured using a waist strap. Raw data from the IMU were processed with a custom MatLab (Mathworks, Inc., Natick, MA) script using the three axis accelerometer data (X = medial-lateral, ML; Y = vertical; and Z = anterior-posterior, AP) and the three axis gyroscope data (X = sagittal rotation, Y = transverse rotation, Z = frontal rotation; all relative to the IMU). A resultant vector was calculated from the corresponding X-Y-Z vectors from each sensor (accelerometer and gyroscope). The root mean squared (RMS) was calculated for the accelerometer (aRMS) and gyroscope (gRMS) for each orthogonal plane and the resultant vector for 3 s following landing.²⁶ Landing was defined as the timepoint corresponding to the peak vertical acceleration.^{25,27} IMU based acceleration metrics have demonstrated acceptable test-retest reliability with intraclass correlation coefficient (ICC) values ranging from 0.73 to 0.90.²⁵ Our laboratory has established the test-retest reliability of gyroscope-based RMS during this task with ICCs ranging from 0.76 to 0.95.

Table 1
NIHTB assessment descriptions.

Assessment	Measurement	Description	Scoring
Flanker Inhibitory Control and Attention Test (FICA)	Executive function; inhibitory control and attention	A central arrow facing left or right is flanked by two arrows on either side, pointing the same or opposite direction as the central arrow. The individual identifies the direction of the central arrow as quickly as possible.	Response time and accuracy.
Dimensional Change Card Sort Test (DCCS)	Executive function; cognitive flexibility	Two objects are presented side-by-side, different in shape and color. The participant matches either the shape or color of the two objects to a third object as quickly as possible.	Response time and accuracy.
Pattern Comparison Processing Speed Test (PCPS)	Cognitive processing speed	The participant indicates if two objects which appear side-by-side are the same or different as quickly as possible.	Response time and accuracy.
Picture Sequence Memory Test (PSM)	Episodic memory	A series of related images appear in a specific order one at a time. After the last picture appears, all of the images are scrambled in the center of the screen in a random order and the participant is tasked with placing the images back in the original display sequence.	Accuracy of the picture placement.

Group differences in demographic variables (age, height, mass, sex), NIHTB composite score (LP, MP, HP), and IMU based measures (aRMS-ML, aRMS-AP, aRMS-vertical, aRMS-resultant, gRMS-sagittal, gRMS-transverse, gRMS-frontal, and gRMS-resultant) were analyzed with one-way ANOVAs. Left and right limbs were initially analyzed independently, but after paired *t*-test analysis showed no difference between limbs ($p = 0.286$ – 0.925) they were averaged together.¹⁸ In the event of a group main effect, post-hoc Bonferroni pairwise comparisons were performed. Corresponding Cohen's *d* effect sizes were calculated and interpreted as weak (0.20), moderate (0.50), and strong (0.80).²⁸ We selected this analysis strategy to mirror the previous literature,^{11,12,18} however, to examine cognition as a continuous variable we also performed Pearson correlation between composite neurocognitive scores and the IMU measures of postural stability. Correlation coefficients (*r*) were interpreted as weak (0.01–0.39), moderate (0.40–0.69), and strong (0.70–1.00). All tests were performed with an a-priori alpha level of $\alpha = 0.05$ using SPSS (Version 26, IBM Corp, Armonk, NY).

3. Results

No differences were identified in age, height, or weight between groups ($p > 0.05$) (Table 2). Significant group differences were identified the aRMS-vertical ($p = 0.043$). Post hoc tests determined the LP group had significantly higher aRMS-vertical ($p = 0.013$, $d = -0.85$) compared to the HP group. Significant group differences were also identified for the aRMS-AP ($p = 0.016$) (Fig. 1). Post hoc tests revealed the LP had significantly lower aRMS-AP ($p = 0.005$, $d = 0.95$) compared to the HP group. No other differences were identified (Table 3, Supplement Figs. 2–4).

A significant moderate, positive correlation was identified between neurocognition and aRMS-AP ($r = 0.430$, $p = 0.003$). A significant weak, negative correlation was identified between neurocognition and aRMS-vertical ($r = -0.331$, $p = 0.026$). The correlation between neurocognition and all other dynamic stability variables was not statistically significant ($p > 0.052$).

4. Discussion

The purpose of this study was to compare dynamic postural stability across collegiate athletes with different levels of neurocognitive performance. Though we did not identify group differences in rotational velocity or the resultant vector, group differences were identified in acceleration in the orthogonal planes during the hop-to-stabilization task. Specifically, greater aRMS-AP and lower aRMS-vertical were exhibited by the HP group when compared to the LP group. Although our hypothesis was not completely supported, these findings suggest that the HP and LP groups may be utilizing different postural stability strategies in response to the hop-to-stabilization task. When considering the nature of this task (i.e. completing a forward hop over a hurdle), it appears that the HP group is moving more efficiently by showing greater acceleration in the AP axis. These findings are consistent with, and expand upon, previous research that have identified biomechanical differences during sport-related tasks when comparing individuals based on neurocognitive performance.^{10–12} Therefore, neurocognitive performance may be related to subtle differences in an athlete's ability to manage postural sway acceleration when attempting to mitigate the effects of the perturbation created from landing.

The central nervous system utilizes multiple degrees of freedom associated with segments and joints along the kinematic chain to maintain center of mass stability during postural tasks.²⁹ It would be expected for there to be a considerable amount of acceleration in the AP axis to achieve postural stability in the forward hop-to-stabilization task used in this study. The greater acceleration of the HP groups in the AP axis could reflect a combination of sagittal plane trunk, hip, and ankle flexion strategies to mitigate landing forces. The increased acceleration in the vertical axis, demonstrated by the LP group, could signify a more ridged landing strategy. The more ridged landing strategy could predispose athletes to future lower extremity injuries while an increased flexion strategy could provide a protective mechanism.³⁰ Although collegiate athletes with lower baseline neurocognitive performance have exhibited greater lower extremity injury risk,⁸ the role of altered biomechanics as a mediating factor has not been prospectively studied.

Table 2
Participant demographics.

	Low performer	Moderate performer	High performer	Group main effect (P)
Sample size (#)	15	14	16	
Age (y)	20.07 ± 1.22	19.50 ± 1.56	19.50 ± 1.71	0.503
Height (cm)	168.33 ± 13.72	174.17 ± 14.35	171.09 ± 12.34	0.510
Weight (kg)	66.77 ± 18.54	69.79 ± 14.05	65.87 ± 15.52	0.802
Sex (male/female)	7/8	8/6	6/10	
Sport (soccer/cheer)	5/10	10/4	10/6	
Previous concussion (#)	6	3	3	
NIHTB composite score	183.73 ± 16.54*†	220.00 ± 6.53*	251.75 ± 16.04	≤0.001

* Significantly different from high performer group.

† Significantly different from the moderate performer group.

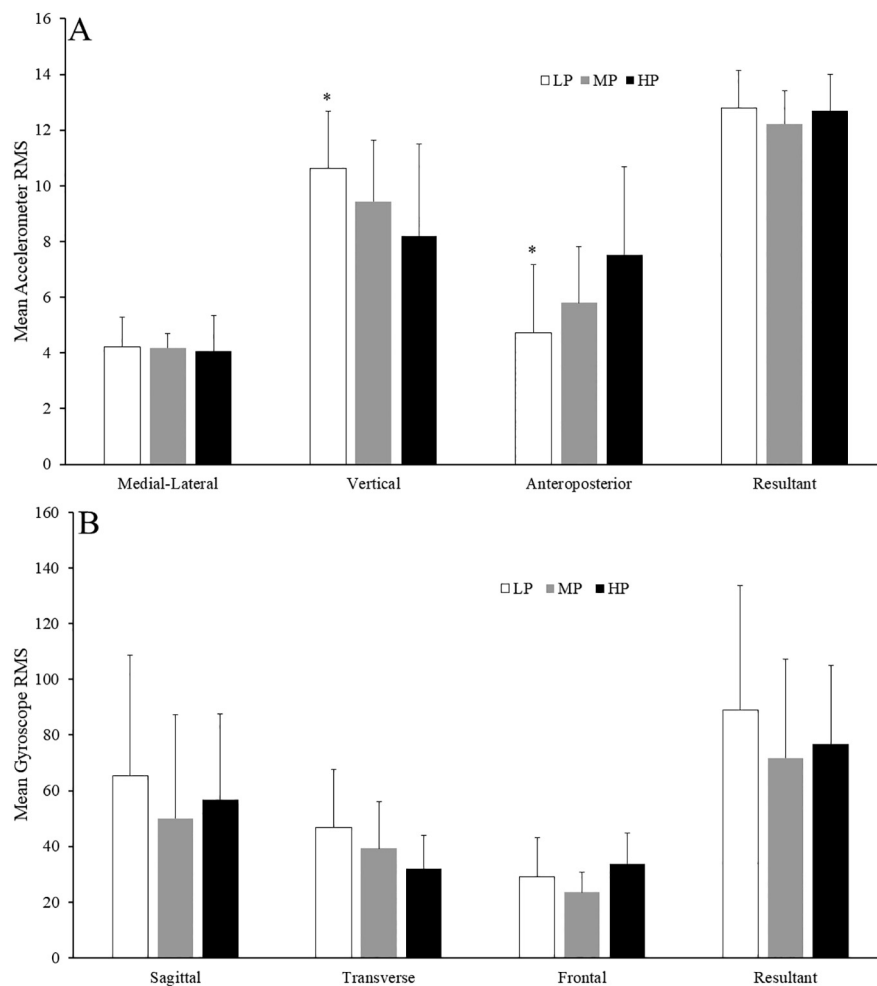


Fig. 1. Bar charts comparing the accelerometer (A) and gyroscope (B) data between groups.

*Significantly different than high performers.

Abbreviations: RMS, root mean square; LP, low performer; MP, moderate performer; HP, high performer.

The results of the current study support previous research that examined the relationship between neurocognition and biomechanics. Herman et al.¹¹ and Shibata et al.¹² identified that healthy adults in the poorer neurocognitive groups exhibited biomechanical patterns consistent with increased lower extremity injury risk based on kinematics and kinetic data captured with 3-dimensional motion capture systems and force plates during athletic tasks. Herman et al.¹¹ reported that the LP group demonstrated 31% higher peak vertical ground reaction force, 26% higher peak proximal anterior shear force, 15% higher external knee abduction moment, and 4.8° greater knee abduction angle compared to the HP group in response to an unanticipated drop-jump landing task. Shibata et al.¹² failed to identify statistically significant differences in kinematics and kinetics during an unanticipated cutting motion across neurocognitive groups. However, further examination of effect sizes revealed the LP group demonstrated higher external knee abduction moment ($d = 0.77$), and lower knee flexion ($d = 0.90$) and hip flexion angles ($d = 0.96$).³¹ These moderate-to-large effects indicate the sample sizes ($n = 7$ – 8 per group) were too small to detect relevant group differences. In the current study, participants in the HP group showed more acceleration in the AP axis which may be valuable for reducing factors commonly associated with joint injury such as increased ground reaction force, anterior shear force, external knee abduction moment, knee abduction angle, decreased knee/hip flexion angle.^{11,12} Integrating 3-dimensional motion capture technology with IMU measures in future studies may enhance the ability to interpret differences in acceleration profiles.

In previous research, the tests utilized and method of establishing groups has varied. Our method of grouping was based off of the composite score from multiple tests and the associated percentile ranks of our participants. While similar methods have been utilized in previous research,^{11,12} future research should explore contextually relevant cutoff values for neurocognitive tests that provide insight into musculoskeletal injury risk to enhance future study designs. Additionally, previous research comparing lower extremity biomechanics in athletes with different levels of neurocognitive performance often omit individuals with neurocognitive scores between the HP and LP groups.^{11,12} Omitting the moderate performance group does allow for clearer separation between individuals considered high and low performers on baseline neurocognitive testing and eliminates participants who have scores encroaching upon the designated thresholds. In the current study, we opted to include all participants which resulted in the inclusion of the MP group which may further contextualize the relationship between neurocognitive performance and postural stability while still providing similar comparisons to previous studies. Although neurocognitive performance was significantly different between each group, relatively few differences in postural stability were identified in the MP group when compared to either the LP or HP groups. This suggests that there may be differences in motor coordination when comparing athletes from each end of the neurocognitive performance spectrum. However, it cannot be determined if the lack of differences in the MP group were because of a true lack of differences in dynamic postural stability compared to either the LP or HP individuals or because of the grouping method.

The four NIHTB tests utilized in this study focused on measuring three major components of neurocognition: episodic memory, processing speed, and executive functioning. Cortical correlates for said domains of cognition are well established in the literature. Executive functioning is primarily found within the prefrontal cortex, while processing speed is primarily linked to the fronto-parietal networks.^{22,24,32} These various constructs of cognition have aspects of autonomy, yet they are interdependent with overlapping neural processes. This can translate into motor coordination and sports as athletes employ cognition to optimize motor coordination by taking in information from the environment (while unanimously suppressing irrelevant stimuli), rapidly processing that data, and calling on previous experiences to help flexibly adapt to the current situation. Previous research has found that more elite soccer players may exhibit greater episodic memory³³ and executive functioning³⁴ when compared to their non-elite counterparts. Additionally, the current study found that higher cognitive functioning may be linked with better postural stability strategies, though more research is warranted to observe if this would also translate into greater success in sports. These findings highlight the importance of neurocognition in sports and may be a relevant factor to consider in the future.

It is important to note that postural stability is an implicit (unconscious) motor act. Historically, research has shown that the modulation of postural stability happens on a subcortical level and therefore a different area of the brain than the centers of neurocognition that we explored in this study.^{35,36} However, existing literature also indicates that the cerebral cortex augments the subcortical mechanisms driving postural stability.^{35,36} As previously mentioned, the contribution of the cerebral cortex on postural stability can be influenced by the current context of the situation and prior experience, or the individuals ability to formulate the appropriate response based on anticipation of the perturbation.³⁶ When an individual has lower neurocognitive performance, one can infer that it may take away from the ability to aid the subcortical mechanisms that drive postural stability. Therefore, those with higher cognitive capacity may adapt to difficult postural tasks by adopting sway strategies that are more advantageous for postural stability. The current study reveals that the individuals with higher cognitive performance utilized a strategy reflected as increased acceleration in the AP axis and lower acceleration in the vertical axis. This may represent a more favorable and adaptable act of postural stability which might contribute to the idea that areas of cognition may have an effect on overall postural stability and ultimately injury risk.

While neurocognition and postural stability were independently examined in this study, future research should explore tasks that integrate cognitive and motor challenges in a manner more representative of a sports setting. Adding a neurocognitive element to motor tasks may add a level of tactical decision-making that may increase the ecological validity. Several studies have developed assessments which combine neurocognitive elements such as reaction time, inhibitory control, processing speed, and memory with motor tasks such as jumping, sprinting, and multidirectional agility.^{37,38} Integrating a neurocognitive performance component into the hop-to-stabilization task utilized in this study may provide more insight into this relationship and provide a direction for future research.

5. Limitations

While participants were encouraged put forth their best effort on the neurocognitive and hop-to-stabilization tests, we cannot account for a potential lack of effort or fluctuating course of effort throughout assessments. All assessments provided the participant's with practice to familiarize themselves with the procedures. However, future research with a larger sample size may overcome these issues by removing a small percentage of individuals with the lowest neurocognitive or hop-to-stabilization scores. Additionally, while we did not include individuals with an acute

injury, we also did not account for lingering chronic injuries, such as chronic ankle instability or concussion, which may alter findings. Future research should further investigate how a history of lower extremity injury or concussion influences the association between neurocognition and dynamic postural stability. While neurocognition and injury history may represent independent variables influencing dynamic postural stability, it is possible that injuries such as concussion could impact both neurocognition and dynamic postural stability directly. Monfort et al.¹⁰ discussed that deficits in neurocognition following concussion could be linked to altered biomechanics and subsequently increased injury risk. While removing the participants with a self-reported history of concussion ($n = 12$) from our study did not alter the findings, it is possible that additional research may identify the individuals who are considered lower neurocognitive performers to have more extensive injury history and subsequently altered biomechanics and increased injury risk. Also, all testing procedures were conducted in a controlled laboratory setting. While this setting enhances internal validity of the study, future research should investigate the impact of neurocognition in a more contextually relevant sport environment. Furthermore, with the focus on postural stability about the center of mass, only one IMU placed on the trunk was used based on previous research.²⁵ Therefore, we cannot make conclusions in regards to the movement patterns of specific segments. Future studies may expand upon these findings by quantifying other movement patterns during landing tasks.

The current study evaluates the relationship between neurocognition and postural stability. While approaches directed in the vertical axis have been seen to be less efficient and less stable, it cannot be definitively stated that these individuals are more susceptible to injury. Conversely, it is also unknown if it is more advantageous for injury prevention to have an increased postural sway acceleration in the AP axis. It is possible increased AP acceleration could predispose athletes to different injuries. Future studies should utilize a prospective study design with injury tracking to determine if any of the differences in IMU based measures observed in this study are indicators of actual injury risk. Additionally, while different neurocognitive performance groups may present with different postural sway characteristics, it is unknown if neurocognitive training would alter biomechanical landing tactics.

6. Conclusion

This study found that neurocognitive performance may influence dynamic postural stability strategies in athletes. Higher neurocognitive performers may use different approaches to perform difficult postural tasks, by adopting strategies associated with lower vertical and higher anteroposterior acceleration compared to lower neurocognitive performers. This research may have implications for identifying individuals that are at a higher risk of injury, in addition to aiding athletic curriculum to involve training in cognitive control.

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Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Confirmation of ethical compliance

This research was approved by the University of Kentucky IRB (protocol #49872). All subjects provided written informed consent prior to participation.

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References

- Hootman JM, Dick R, Agel J. Epidemiology of collegiate injuries for 15 sports: summary and recommendations for injury prevention initiatives. *J Athl Train* 2007;42(2):311–319.
- Safran MR, Benedetti RS, Bartolozzi RA et al. Lateral ankle sprains: a comprehensive review part 1: etiology, pathoanatomy, histopathogenesis, and diagnosis. *Med Sci Sports Exerc* 1999;31(7 Suppl):S429–S437.
- Smith MA, Smith WT, Kosko P. Anterior cruciate ligament tears: reconstruction and rehabilitation. *Orthop Nurs* 2014;33(1):14–24.
- Anandacoomarasamy A, Barnsley L. Long term outcomes of inversion ankle injuries. *Br J Sports Med* 2005;39(3):e14.
- Lohmander LS, Östenberg A, Englund M et al. High prevalence of knee osteoarthritis, pain, and functional limitations in female soccer players twelve years after anterior cruciate ligament injury. *Arthritis Rheum* 2004;50(10):3145–3152.
- von Porat A, Roos EM, Roos H. High prevalence of osteoarthritis 14 years after an anterior cruciate ligament tear in male soccer players: a study of radiographic and patient relevant outcomes. *Ann Rheum Dis* 2004;63(3):269–273.
- Aaltonen S, Karjalainen H, Heinonen A et al. Prevention of sports injuries: systematic review of randomized controlled trials. *Arch Intern Med* 2007;167(15):1585–1592.
- Swanik C, Covassin T, Stearne D et al. The relationship between neurocognitive function and noncontact anterior cruciate ligament injuries. *Am J Sports Med* 2007;35:943–948.
- Wilkerson GB. Neurocognitive reaction time predicts lower extremity sprains and strains. *Int J Athl Ther Train* 2012;17(6):4–9.
- Monfort SM, Pradarelli JJ, Grooms DR et al. Visual-spatial memory deficits are related to increased knee valgus angle during a sport-specific sidestep cut. *Am J Sports Med* 2019;47(6):1488–1495.
- Herman DC, Barth JT. Drop-jump landing varies with baseline neurocognition: implications for anterior cruciate ligament injury risk and prevention. *Am J Sports Med* 2016;44(9):2347–2353.
- Shibata S, Takemura M, Miyakawa S. The influence of differences in neurocognitive function on lower limb kinematics, kinetics, and muscle activity during an unanticipated cutting motion. *Phys Ther Res* 2018;21(2):44–52.
- Gokeler A, McKeon PO, Hoch MC. Shaping the functional task environment in sports injury rehabilitation: a framework to integrate perceptual-cognitive training in rehabilitation. *Athl Train Sports Health Care* 2020;12(6):283–292.
- Plisky PJ, Rauh MJ, Kaminski TW et al. Star excursion balance test as a predictor of lower extremity injury in high school basketball players. *J Orthop Sports Phys Ther* 2006;36(12):911–919.
- Butler RJ, Lehr ME, Fink ML et al. Dynamic balance performance and noncontact lower extremity injury in college football players: an initial study. *Sports Health* 2013;5(5):417–422.
- Pau M, Porta M, Arippa F et al. Dynamic postural stability, is associated with competitive level, in youth league soccer players. *Phys Ther Sport* 2019;35:36–41.
- Rahnama L, Salavati M, Akhbari B et al. Attentional demands and postural control in athletes with and without functional ankle instability. *J Orthop Sports Phys Ther* 2010;40(3):180–187.
- Talarico MK, Lynall RC, Mauntel TC et al. Static and dynamic single leg postural control performance during dual-task paradigms. *J Sports Sci* 2017;35(11):1118–1124.
- Westwood C, Killelea C, Faherty M et al. Postural stability under dual-task conditions: development of a post-concussion assessment for lower-extremity injury risk. *J Sport Rehabil* 2020;29(1):131.
- Weintraub S, Dikmen SS, Heaton RK et al. The cognition battery of the NIH toolbox for assessment of neurological and behavioral function: validation in an adult sample. *J Int Neuropsychol Soc* 2014;20(6):567–578.
- Weintraub S, Dikmen SS, Heaton RK et al. Cognition assessment using the NIH toolbox. *Neurology* 2013;80(11 Supplement 3):S54.
- Carlozzi NE, Tulsy DS, Chiaravalloti ND et al. NIH toolbox cognitive battery (NIHTB-CB): the NIHTB pattern comparison processing speed test. *J Int Neuropsychol Soc* 2014;20(6):630–641.
- Zelazo PD, Anderson JE, Richler J et al. NIH toolbox cognitive battery (CB): validation of executive function measures in adults. *J Int Neuropsychol Soc* 2014;20(6):620–629.
- Dikmen SS, Bauer PJ, Weintraub S et al. Measuring episodic memory across the lifespan: NIH toolbox picture sequence memory test. *J Int Neuropsychol Soc* 2014;20(6):611–619.
- Heebner NR, Akins JS, Lephart SM et al. Reliability and validity of an accelerometry based measure of static and dynamic postural stability in healthy and active individuals. *Gait Posture* 2015;41(2):535–539.
- Akins JS, Heebner NR, Lovalekar M et al. Reliability and validity of instrumented soccer equipment. *J Appl Biomech* 2015;31(3):195–201.
- Sell TC. An examination, correlation, and comparison of static and dynamic measures of postural stability in healthy, physically active adults. *Phys Ther Sport* 2012;13(2):80–86.
- Cohen J. Statistical power analysis for the behavioral sciences, Hillsdale, N.J., L. Erlbaum Associates, 1988.
- Iqbal K. Mechanisms and models of postural stability and control, 2011 annual international conference of the IEEE engineering in medicine and biology society, 2011. p. 7837–7840.
- Leppänen M, Pasanen K, Kujala UM et al. Stiff landings are associated with increased ACL injury risk in young female basketball and floorball players. *Am J Sports Med* 2017;45(2):386–393.
- Porter KH, Quintana C, Hoch MC. The relationship between neurocognitive function and biomechanics: a critically appraised topic. *J Sport Rehabil* 2020:1–6.
- Diamond A. Executive functions. *Annu Rev Psychol* 2013;64(1):135–168.
- Williams AM, Ward P, Bell-Walker J et al. Perceptual-cognitive expertise, practice history profiles and recall performance in soccer. *Br J Psychol* 2012;103(3):393–411.
- Vestberg T, Gustafson R, Maurel L et al. Executive functions predict the success of top-soccer players. *PLoS ONE* 2012;7(4):e34731.
- Bolton DAE. The role of the cerebral cortex in postural responses to externally induced perturbations. *Neurosci Biobehav Rev* 2015;57:142–155.
- Jacobs JV, Horak FB. Cortical control of postural responses. *J Neural Transm* 2007;114(10):1339.
- Wilke J, Vogel O, Ungricht S. Can we measure perceptual-cognitive function during athletic movement? A framework for and reliability of a sports-related testing battery. *Phys Ther Sport* 2020;43:120–126.
- Millikan N, Grooms DR, Hoffman B et al. The development and reliability of 4 clinical neurocognitive single-leg hop tests: implications for return to activity decision-making. *J Sport Rehabil* 2019;28(5):536–544.